CHAPTER 8:

Future of the Future

How big the hourglass?
How deep the sand?
I shouldn't hope to know, but here I stand.
Hozier, "No Plan"

In 1969, Martin Rees was not yet Astronomer Royal, Lord Rees, Baron of Ludlow. He was a postdoc cosmologist at Cambridge University, thinking about the end of everything, publishing a six-page paper titled "The Collapse of the Universe: An Eschatological Study," which he would later describe as "rather fun." In the introduction, Rees explained that while the observational evidence was still uncertain, it indicated that "the universe is indeed fated to collapse. All structural features of the cosmic scene would be destroyed during this devastating compression." Part of what made the paper fun to Rees was calculating that, in the coming collapse, all the stars will be destroyed by ambient radiation, from the outside in. Who wouldn't enjoy the thought of stars catching fire?

Despite Rees's arguments in favor of a Big Crunch, the data remained ambiguous for decades. Was the universe closed (recollapsing) or open (eternally expanding)? In 1979, Freeman Dyson, at the Institute for Advanced Study in Princeton, decided to explore the other side of the argument, saying, "I shall not discuss the closed universe in detail, since it gives me a feeling of claustrophobia to imagine our whole existence confined within the box." The open universe model was a pleasantly roomy alternative. In his paper "Time Without End: Physics and Biology in an Open Universe," he worked through quantitative predictions of what an open universe might mean for humanity, working out a method by which future beings might, through regulating their activity and entering periods of hibernation, avoid oblivion into the infinite future as the rest of the cosmos dissolves around them.[fn1] While most of the paper consists of calculations and theoretical discussions, the introduction contains some sharp words aimed at the physics mainstream for unfairly disdaining the